

Response to Reviewers

Manuscript #Access-2025-52745
“Topology-Constrained NeuroB-Rep: Contact-Aware CAD
Reconstruction from 3-D Point Clouds”

We sincerely thank both reviewers for their thorough and constructive feedback. Every comment has been addressed in the revised manuscript. Below we respond to each point, describe the specific changes made, and provide cross-references to the relevant sections, tables, and figures in the **revised** manuscript.

Reviewer 1

R1-M1: Clarity of Contribution vs. Point2CAD

Reviewer Comment: *The distinction between the proposed method and Point2CAD is not sufficiently clear. A side-by-side comparison table or flowchart would help readers understand the novelty.*

Author Response: We have added Table 2 (Pipeline Comparison), which contrasts NeuroB-Rep and Point2CAD across seven design dimensions (topology handling, contact objective, gating schedule, continuity enforcement, freeform surfaces, export format, and evaluation protocol). The Introduction (Sec. I) has been restructured so that the industrial motivation paragraph precedes the contribution list, making the positioning against Point2CAD explicit from the outset.

Changes Made: New Table 2 in Sec. IV; Introduction restructured (Sec. I, paragraphs 3–5).

R1-M2: Evaluation Protocol Consistency

Reviewer Comment: *The evaluation protocol mixes different scales and metrics. A rationale for this hybrid approach is needed.*

Author Response: We have added a dedicated “Rationale for Hybrid Evaluation” paragraph in Sec. V-B explaining why contact-gap energies are evaluated in a normalized scale while geometric distances are reported in millimeters: contact tolerances are scale-invariant design intents, whereas Chamfer/Hausdorff must be in physical units to be interpretable by manufacturing engineers. We also clarify the ICP alignment step and its role in bridging the two scales.

Changes Made: New paragraph “Rationale for Hybrid Evaluation” in Sec. V-B.

R1-M3: Missing Baseline Comparison

Reviewer Comment: *Table 3 reports only your method. At least one baseline (e.g., Point2CAD) should be included under the same protocol.*

Author Response: Table 7 (Geometry Consistency after ICP) now includes a Point2CAD column. We ran Point2CAD on the same three components (Impeller, Shaft, Casing) under the identical ICP protocol with 120 independent repeats per component. NeuroB-Rep consistently outperforms Point2CAD: geometric accuracy and surface coincidence errors are 2–3× lower, while length errors show 4–8× improvement. (Note: The table was Table 3 in the original submission; it is now Table 7 due to newly added tables.)

Changes Made: Table 7 extended with Point2CAD column (Sec. VI-G).

R1-M4: Failure Cases

Reviewer Comment: *The paper does not discuss failure modes. Please analyze cases where the method does not perform well.*

Author Response: We have added Sec. VIII-E “Failure Modes and Mitigations” analyzing the cross-boundary misassignment observed in Figs. 5–6. The primary failure originates in the upstream PointNet segmentation stage when geometrically similar adjacent surfaces receive the same label. We describe the downstream effects (degraded data-term fit, missing contact edges) and propose three mitigations: adaptive layer thresholds, boundary-aware resampling, and multi-scale segmentation.

Changes Made: New Sec. VIII-E with failure analysis and three mitigation strategies.

R1-m1: Abstract Wording

Reviewer Comment: *“0.01 mm” in the abstract should reflect the range across components.*

Author Response: Changed to “0.01–0.03 mm” in the abstract and corresponding results text.

Changes Made: Abstract and Sec. VI updated.

R1-m2: Formula Notation

Reviewer Comment: *The relationship between $d_{ij,k}$ and $s_{ij,k}$ after Eq. (2) is unclear.*

Author Response: Added a clarifying sentence after Eq. (2) linking $d_{ij,k}$ (gap distance) and $s_{ij,k}$ (projected overlap distance) and explaining their complementary roles in the contact objective.

Changes Made: Sentence added after Eq. (2) in Sec. IV-B.

R1-m3: Figure Reference Order

Reviewer Comment: *Figures are not referenced in sequential order.*

Author Response: All figure references have been checked and corrected to follow sequential order throughout the manuscript.

Changes Made: Cross-references updated throughout.

Reviewer 2

R2-C1: Comparison with Six Cited Methods

Reviewer Comment: *The paper should compare with Li et al. 2019, ComplexGen, PrimitiveNet, Supervised Fitting, ParSeNet, and BrepGen, not just cite them.*

Author Response: We provide two levels of comparison: (1) Table 1 (Feature Comparison) gives a qualitative comparison across all surveyed methods on seven capability axes (topology support, freeform surfaces, STEP export, etc.). (2) Table 7 provides a full quantitative comparison with Point2CAD under identical protocol. A new Sec. VIII-B (“Scope of Baseline Comparisons”) explains why the remaining six methods—Supervised Fitting, PrimitiveNet, ParSeNet, ComplexGen, BrepGen, and Maurell et al.—cannot be compared numerically: dataset mismatch (ABC/ShapeNet/synthetic vs. industrial scans), output-format disparity (chain complexes, latent codes vs. STEP), and evaluation-metric incompatibility (IoU/F-score vs. mm-locked Chamfer/Hausdorff + contact-gap). Point2CAD is the only candidate sharing the same pipeline architecture, publicly available code, and compatible input format.

Changes Made: New Table 1 (Sec. III); extended Table 7 (Sec. VI-G); new Sec. VIII-B explaining comparison scope.

R2-C2: Paper Outline Missing

Reviewer Comment: *Add a paper outline at the end of the Introduction.*

Author Response: A paper-outline paragraph has been added at the end of Sec. I, listing all sections and their content.

Changes Made: Outline paragraph added at end of Sec. I.

R2-C3: Motivation Paragraph Position

Reviewer Comment: *The industrial motivation should appear before the contribution list, not after.*

Author Response: The motivation paragraph has been moved to precede the contribution list in Sec. I.

Changes Made: Paragraph reordered in Sec. I.

R2-C4 & R2-C5: Related Work Structure and Depth

Reviewer Comment: *The Related Work section has a single subsection and lacks detailed positioning against existing methods.*

Author Response: Sec. III (Related Work) has been restructured into four subsections: (A) Point-Cloud Segmentation, (B) Primitive Fitting and Assembly, (C) B-Rep Learning and Generative Models, and (D) Hybrid Reconstruction and Point2CAD. Each subsection discusses the strengths and limitations of prior methods and explicitly positions our approach. Table 1 provides a structured feature comparison. New references have been added: Li et al. 2019, PrimitiveNet (Huang 2021), Maurell et al. 2024.

Changes Made: Sec. III restructured with four subsections and Table 1.

R2-C6: Face/Edge/Vertex Recovery from Points

Reviewer Comment: *How are faces, edges, and vertices recovered from the unstructured point cloud?*

Author Response: Sec. IV-A now includes a dedicated paragraph (“Patch Graph Construction”) explaining the kNN co-label adjacency mechanism: for each point, the k nearest neighbors are queried; if a neighbor carries a different patch label, an edge is recorded between the two patches. This yields the face adjacency graph whose edges define candidate contact boundaries.

Changes Made: New paragraph in Sec. IV-A-2 (“From Points to a Patch Graph”).

R2-C7: Single Subsection in Method

Reviewer Comment: *Section III-A contains only one subsection, which violates standard structuring conventions.*

Author Response: The Method section (now Sec. IV after the insertion of a new Preliminaries section) has been expanded with multiple subsections (IV-A through IV-F), resolving the single-subsection issue. The original Related Work (now Sec. III) was similarly restructured into four subsections (III-A through III-D).

Changes Made: Sec. IV-A now contains two subsubsections (IV-A-1 and IV-A-2), resolving the single-subsection issue; Sec. III (Related Work) restructured into four subsections (A–D).

R2-C8: Jang et al. Explanation

Reviewer Comment: *The description of Jang et al. [12] is too brief. Expand the explanation of the mesh repair principles.*

Author Response: Sec. II-C now provides a three-step description of Jang et al.’s method: (i) detection of interpenetrating triangles via oriented bounding-box pre-filtering, (ii) computation of exact intersection curves between conflicting triangles, and (iii) local re-meshing by splitting edges along intersection curves and re-triangulating affected patches. We note the method operates in a single pass without global re-meshing, making it suitable as an online diagnostic.

Changes Made: Sec. II-C expanded from one sentence to a full paragraph.

R2-C9: Projection Operator Clarity

Reviewer Comment: *The passage describing the projection operator is hard to follow.*

Author Response: The projection operator description has been rewritten for clarity, with explicit notation for the nearest-point projection onto each parametric surface and the role of the projection in computing gap distances.

Changes Made: Projection passage rewritten in Sec. IV-B.

R2-C10: “Soft Barrier” Definition

Reviewer Comment: *Define “soft barrier” at first use.*

Author Response: The term is now defined in Sec. II-C (Preliminaries) at its first occurrence: “a smooth penalty function $\log(1 + \exp(\tau - s))$ that grows rapidly as the distance s between projected points on two patches decreases below threshold τ .”

Changes Made: Definition added in Sec. II-C.

R2-C11: Inconsistent Paragraph Labels

Reviewer Comment: *The “a:” style paragraph labels are inconsistent.*

Author Response: All paragraph labels have been unified using the standard `\paragraph{}` command throughout the manuscript.

Changes Made: Paragraph labels standardized throughout.

R2-C12: Fig. 3 Panel Count

Reviewer Comment: *The caption references panels (a,b,c,d) but only three panels are visible.*

Author Response: This was a caption error. The original figure has always contained three panels. The caption has been corrected to reference (a), (b), and (c) only.

Changes Made: Fig. 3 caption corrected to three panels.

R2-C13: PCA Normalization Justification

Reviewer Comment: *The PCA canonical transform is used without justification. An ablation would strengthen the claim.*

Author Response: We conducted a PCA ablation experiment (5 seeds, 60 iterations each) comparing convergence with and without PCA normalization. Results are reported in Table 6 (Sec. VI-B). PCA normalization yields a modest but consistent improvement: Chamfer distance improves by 0.3% and Hausdorff by 1.4%, with lower variance across seeds. The primary benefit is stabilizing the initial fit and reducing sensitivity to scan orientation.

Changes Made: New Table 6 and Sec. VI-B “Effect of PCA Canonical Transform.”

R2-C14: ISO 5459 Datum Explanation

Reviewer Comment: *The reference to ISO 5459 needs more context.*

Author Response: Sec. V-A now includes an expanded explanation of the ISO 5459 datum framework: how datum features are established from physical surfaces, and how this relates to our evaluation’s reference geometry and coordinate alignment via ICP.

Changes Made: Expanded explanation in Sec. V-A.

R2-C15: GPU Implementation Details

Reviewer Comment: *No information is provided about the GPU implementation.*

Author Response: We have added implementation details in Sec. V-E (Computational Cost) and Table 4: all optimizations run on a single NVIDIA RTX 3090 using PyTorch autograd with CUDA-batched energy evaluations. NeuroB-Rep requires ~ 7 min / 4.8 GB per component vs. Point2CAD’s ~ 4 min / 3.2 GB. The overhead stems from the contact-gap and continuity terms, which require pairwise boundary-point queries absent in Point2CAD’s distance-only objective.

Changes Made: New Sec. V-E and Table 4.

R2-C16: Artifact URL

Reviewer Comment: *Provide a public artifact URL for reproducibility.*

Author Response: The solver code, evaluation scripts, and default configuration files are now publicly available at <https://github.com/JuneWooKang/TopoConstrained-NeuroBrep>. Due to proprietary constraints, the industrial scan data cannot be included; however, all scripts reproduce the reported metrics when provided with compatible point-cloud inputs.

Changes Made: URL added in Sec. VII.

R2-C17: Table 3 Baseline Column

Reviewer Comment: *Table 3 should include at least one baseline for comparison.*

Author Response: The consistency table (now Table 7 in the revised manuscript) includes a Point2CAD column evaluated under the identical ICP protocol (120 repeats per component). See R1-M3 above for details.

Changes Made: Point2CAD column added to Table 7 (Sec. VI-G).

R2-C18: “Reproducibility Hygiene” Wording

Reviewer Comment: *The term “reproducibility hygiene” is informal.*

Author Response: Changed to “reproducibility protocol” throughout the manuscript.

Changes Made: Terminology updated in Sec. VII.

Additional Corrections

During revision, we identified and corrected the following issues:

- **Iteration count (1,000 $\rightarrow$ 120):** The original submission stated “1,000 independent repeats” in the evaluation description. This was a typographical error. All experiments were conducted with 120 independent repeats per component, as correctly reflected in the Table 7 caption. The text has been corrected throughout.
- **Fig. 3 caption (4 $\rightarrow$ 3 panels):** The original image has always contained three panels. The caption erroneously referenced four panels (a,b,c,d). The caption now correctly references panels (a), (b), (c).
- **Abstract grammar:** “a based on temperature gating mechanism” corrected to “a temperature-based gating mechanism.”
- **Computational cost reporting:** Added Table 4 reporting training time, GPU memory, and per-component overhead for both NeuroB-Rep and Point2CAD on an NVIDIA RTX 3090.
- **New Acknowledgment section:** Added NRF funding acknowledgment before the References.
- **Section renumbering:** A new Sec. II (Preliminaries) was inserted, shifting all subsequent section numbers by one. Correspondingly, four new tables (Tables 1, 2, 4, 6) were added, renumbering existing tables.

We believe these revisions comprehensively address all reviewer concerns. We are grateful for the opportunity to strengthen the manuscript and welcome any further suggestions.